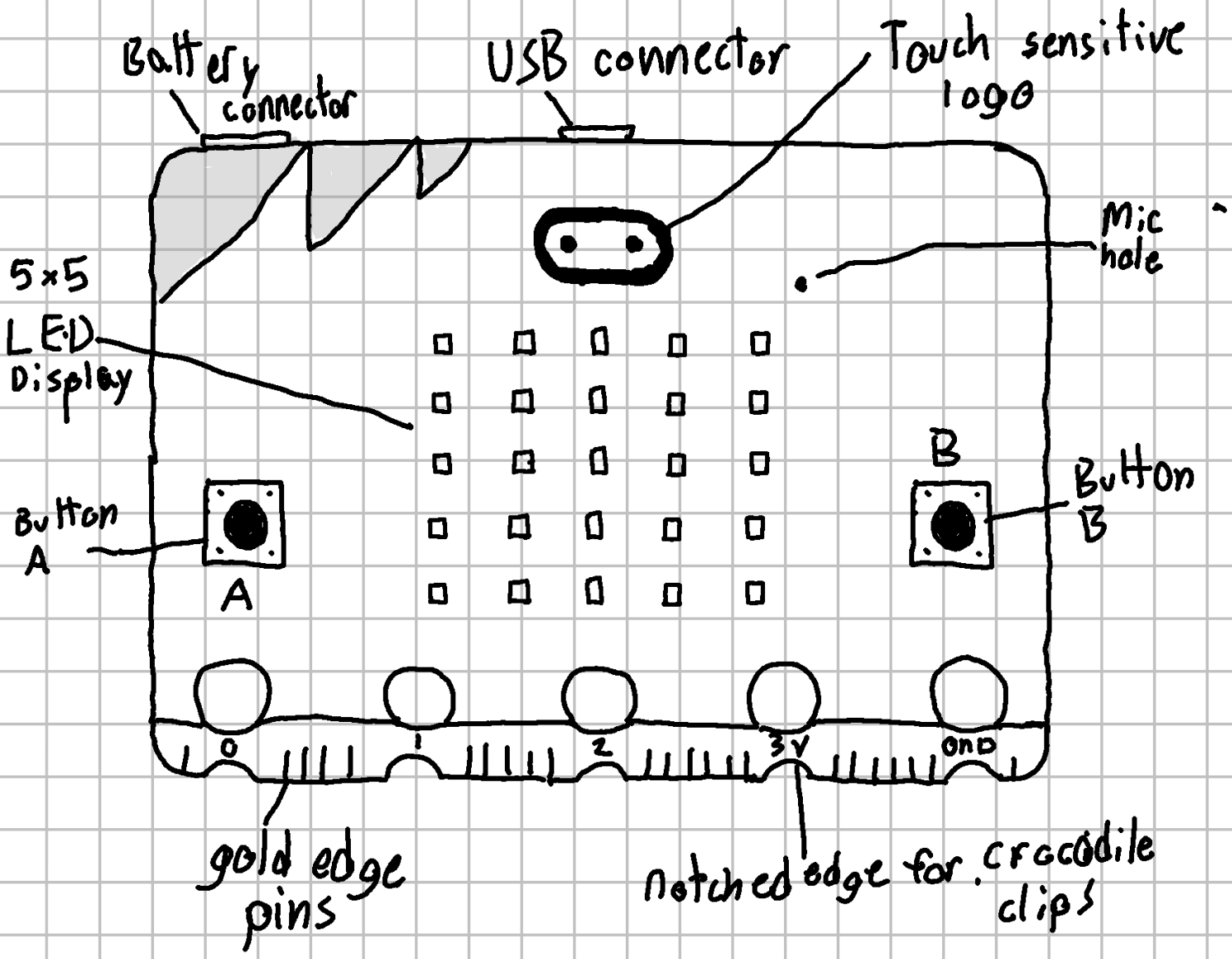
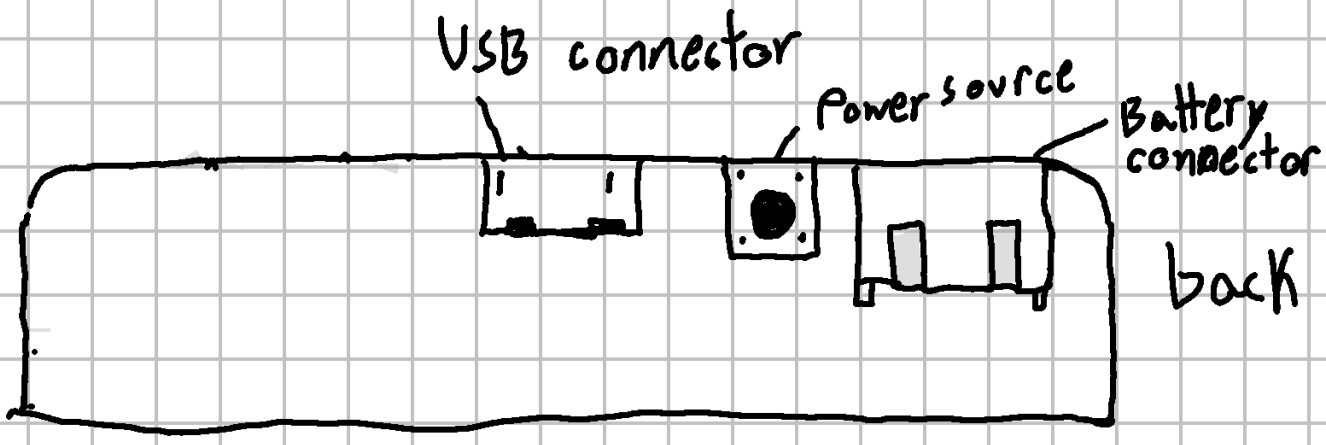
Micro:bit


Battery connector
USB connector
Touch sensitive logo
Mic hole
5x5 LED Display
Button A
A
B
Button B
0
1
2
3V
GND
gold edge pins
notched edge for crocodile clips


USB connector
Power source
Battery connector
back

Ω resistors - how to tell the difference between those in the kit

220 : 2 skinny black lines

1K : 2 thick black lines

10K : 1 thick red line

GOLDEN RULES

- Always use the 220Ω with LEDs
- micro:bit runs at 3.3v, never connect to a 5v.
- Breadboard is reusable, pull components out by the body not the legs.
- When in doubt double check your wiring before downloading or rewriting the code.

Curious Components

- 1 digit LED display: Tally points in a game.
- Thermistor: when you breathe on it something happens
- servo motor: rotates an arm to hit you when a button is pressed.

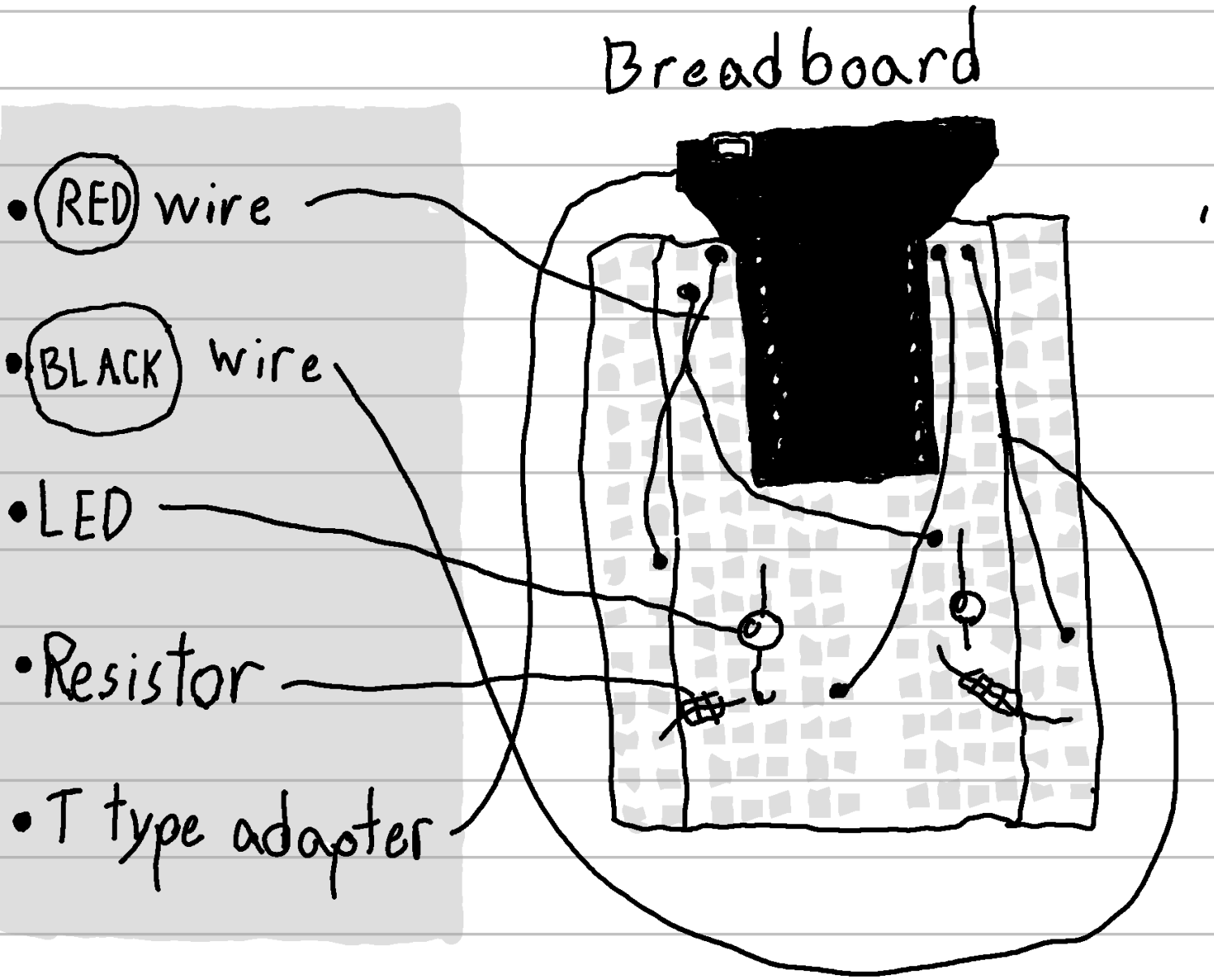
hello w🌍rld

what does the show string block do?
- The show string prints a scrolling message along the 5x5 LED grid.

Difference between on start and forever?
- on start does it once and forever repeats it forever.

What would you want it to say and why?
- PP1, since its the name of my project.

My first Breadboard



Microbit react

My Microbit has 3 different inputs,

- Button A
- Button B
- Touch sensitive logo

Event driven programming: the code only runs when something happens

Accelerometer: Sensor that detects motion and orientation
Ex. Air bag, mouse, phone.

FARM HOUSE IDEAS

Servo door lock: Opens and closes a door with a button press.

Materials:

- Servo motor
- Jumper wires

Code:

When button A is pressed it sets P0 to 1, when B is pressed it sets P0 to 0.

- Servodoors,
- togglable lights
- motion porch light
- fans
- DOORBELL
- intruder alarm
- Servo blinds (light receptive)